

The background is a dark blue gradient. On the left side, there are faint, light blue circuit-like lines and small circles. On the right side, there are two large, overlapping circles in shades of blue and teal.

FIT1047

ASSIGNMENT 4

TASK 3

HARITHIRAA YOGARAJ

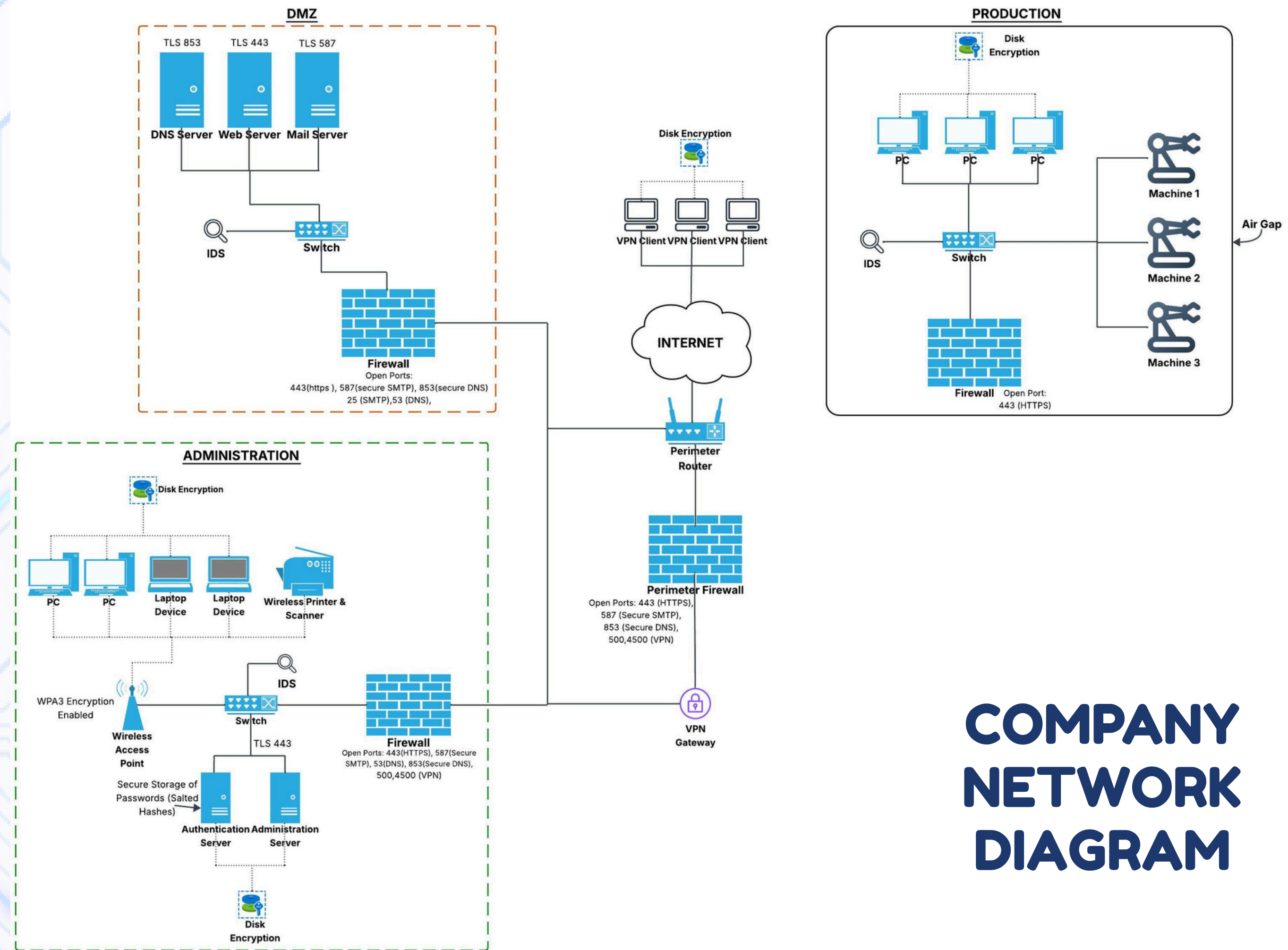
SECURE NETWORK INFRASTRUCTURE DESIGN

DMZ, ADMINISTRATION, PRODUCTION ZONES
WITH VPN AND FIREWALL CONTROLS



Overall Structure

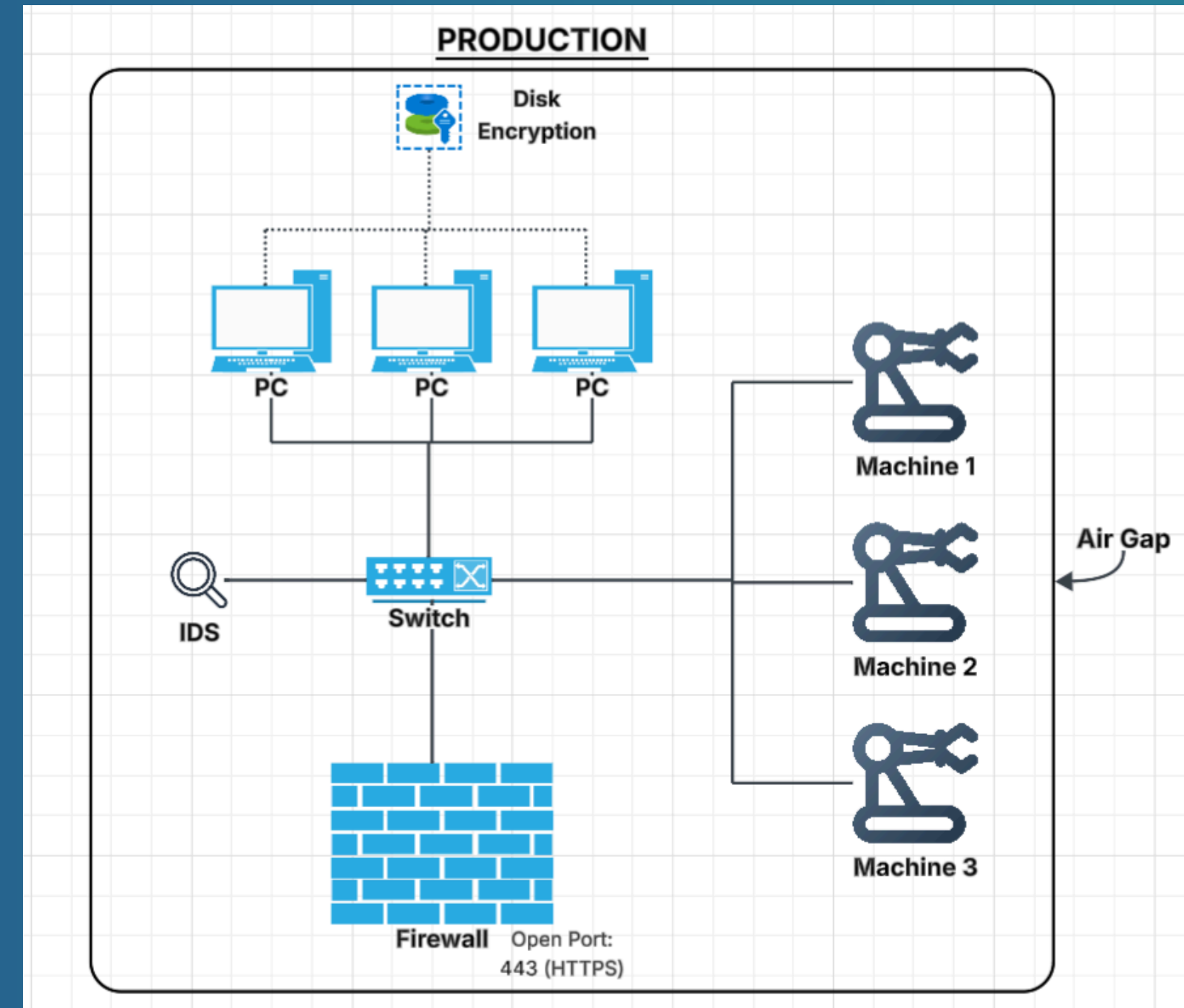
- The network diagram I have designed is structured into three segment zones for security and efficiency.
- DMZ Zone – Hosts public-facing servers such as the web, mail and DNS servers and adds an extra layer of security by isolating them from the internal network.
- Administration Zone – Hosts internal devices like admin PCs, server, printers, and authentication systems. It manages and controls internal operations and access.
- Production Zone – Contains critical systems like machinery and production PCs. It's isolated (air-gapped) for maximum security to protect core industrial operations



Production Zone



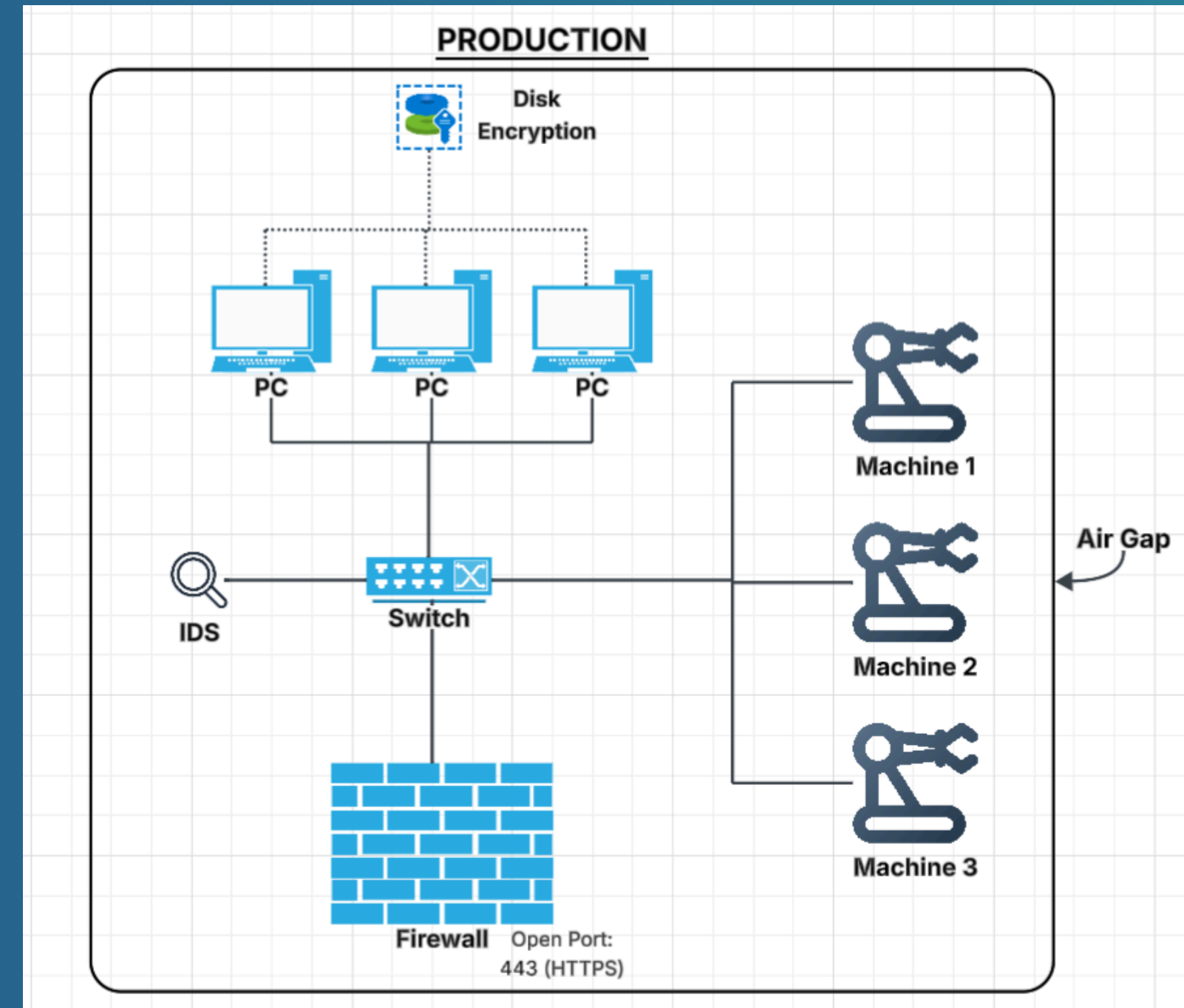
- In the Production zone, all the production PCs and machineries are placed with tight security.
- Entities in this zone:
 - Production PCs – Used by staff to monitor, control or interact with the machinery during manufacturing processes, ensuring operations run smoothly and efficiently.
 - Machinery – automates the manufacturing process of vacuum cleaners and operates continuously 24/7 to ensure high efficiency, consistent quality and uninterrupted production.
 - Switch – Interconnects production devices allowing them to communicate internally and work together efficiently in the production zone.



Production Zone



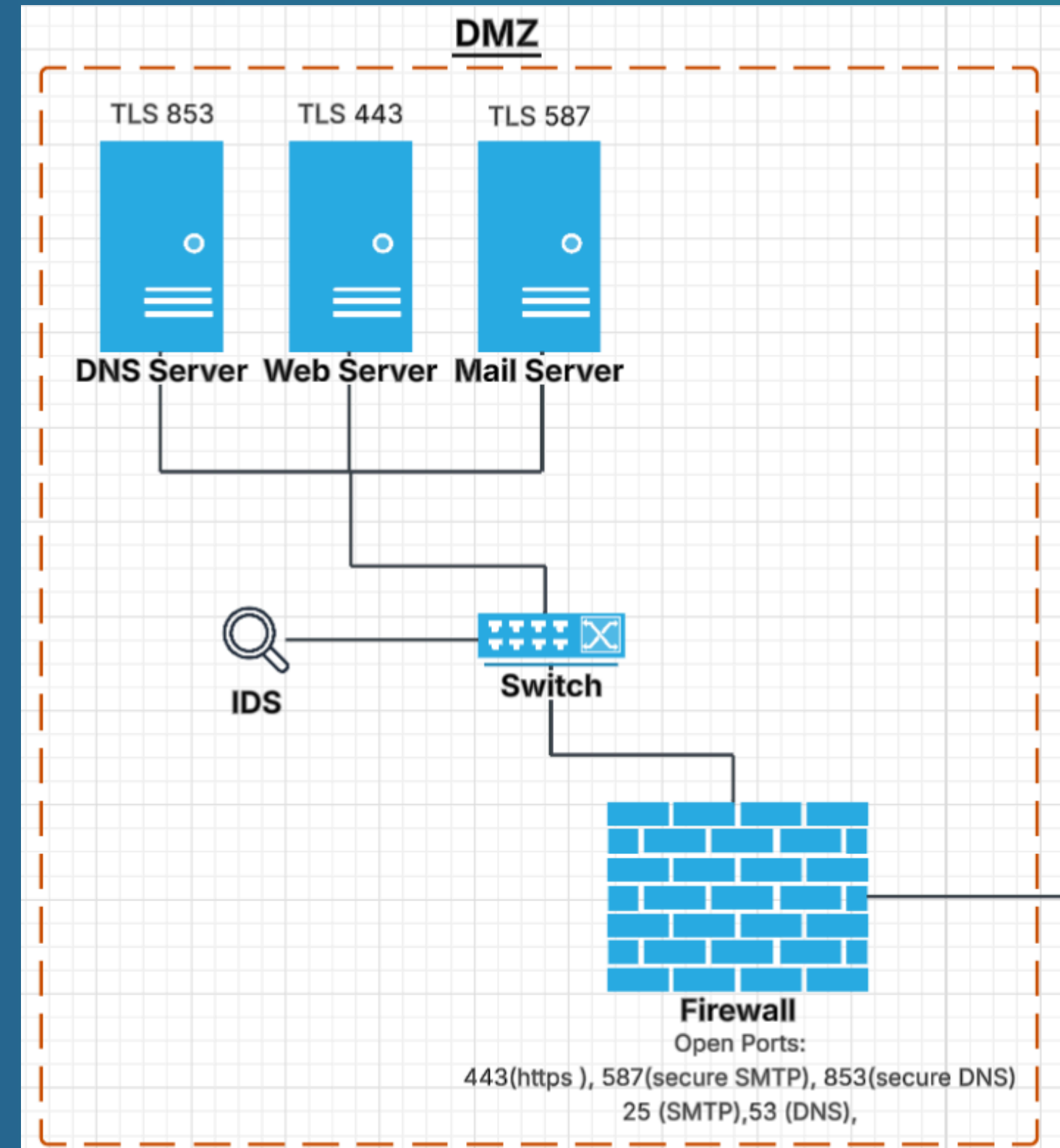
- Security Controls in the production zone:
 - IDS – Monitors internal traffic for malicious behavior or internal threats helping to identify potential security threats and protect the system from cyberattacks.
 - Air Gap – This zone is isolated from other zones to ensure stable operations and secure data transfer, reducing the risk of downtime or external interference.
 - Firewall – Controls traffic in and out of the production zone. It only allows very limited internal connections and blocks all direct internet traffic since it's air-gapped.
 - Disk Encryption – Helps protect important production data from being accessed or leaked.



DMZ Zone



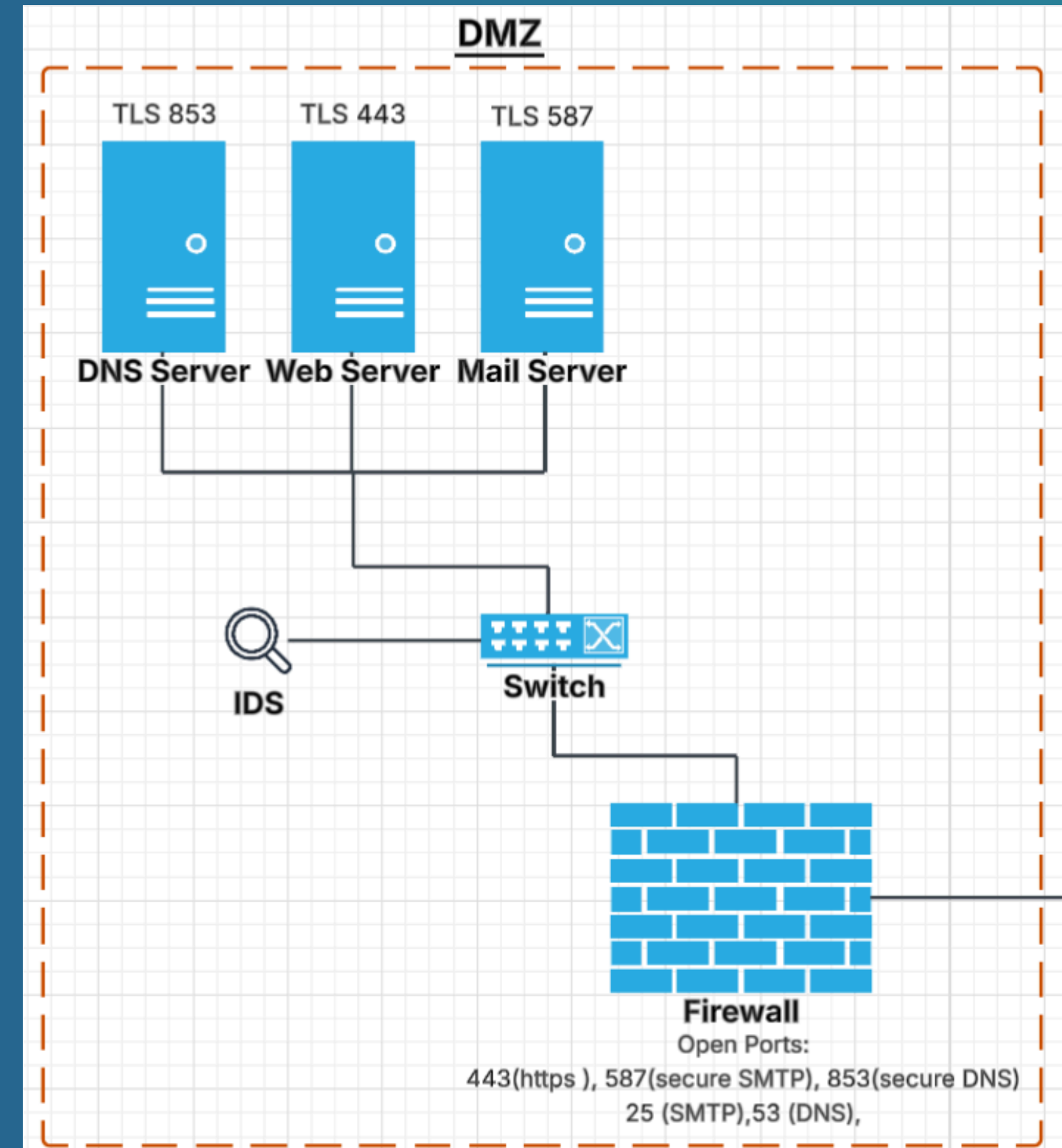
- The DMZ hosts public-facing services that interact with the external internet, while isolating them from internal networks for added security.
- Entities in this zone:
 - Web Server – Hosts the company website, making it accessible to users over the internet via secure HTTPS (port 443), while being isolated from internal systems for security.
 - Mail Server – Sends and receives company emails using ports 25 and 587.
 - DNS Server – Looks up domain names; supports both standard DNS (port 53) and DNS over TLS (port 853) for secure queries.
 - Switch – Connects DMZ devices allowing them to communicate internally and manages internal traffic flow.



DMZ Zone



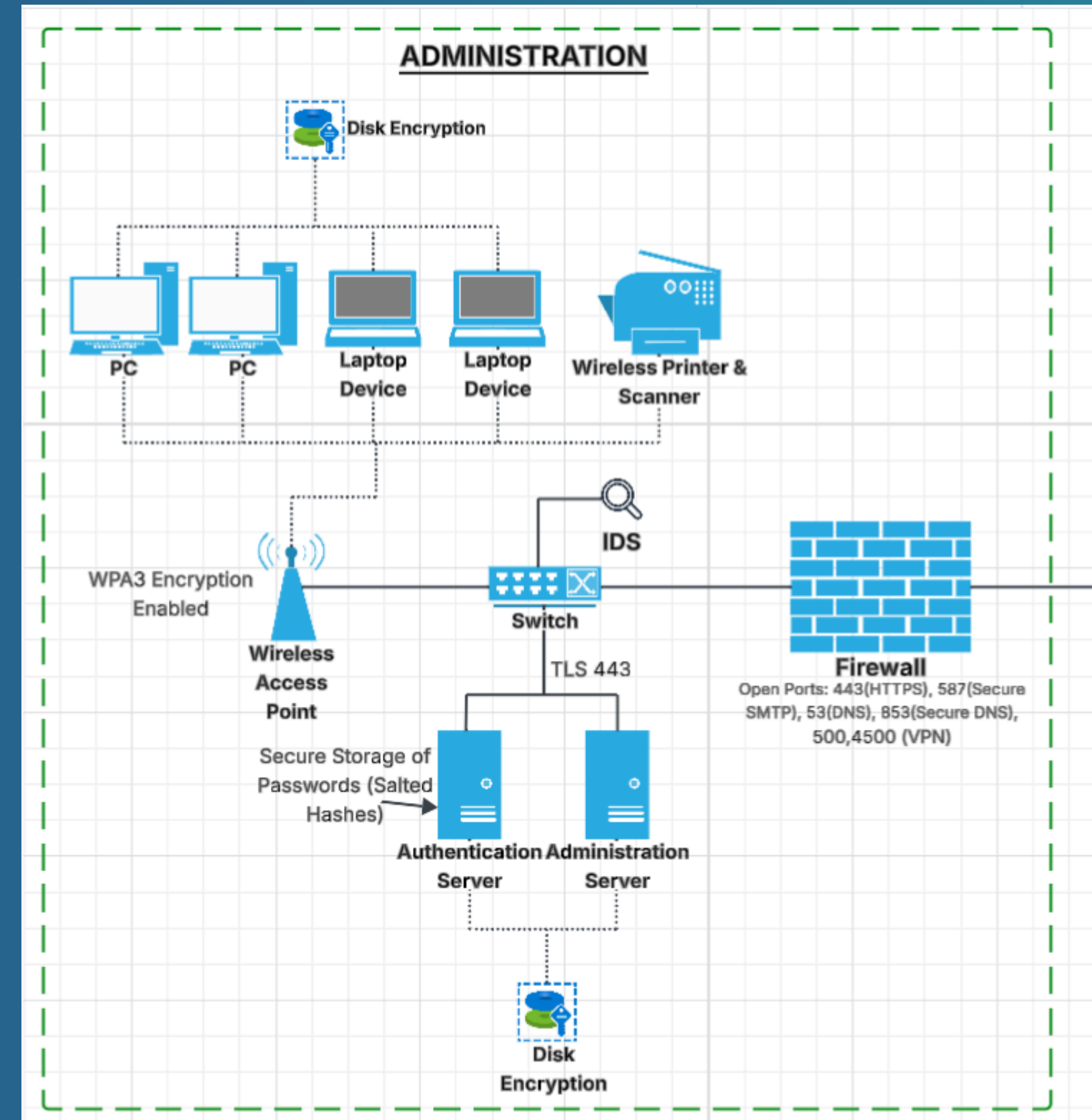
- Security Controls in the DMZ Zone:
 - IDS – Monitors incoming and outgoing traffic to detect any signs of attacks targeting public-facing services like the web, mail, or DNS servers.
 - Firewall – Controls and filters traffic between the DMZ and other zones to protect internal systems while allowing limited external access to public-facing servers.



Administration Zone



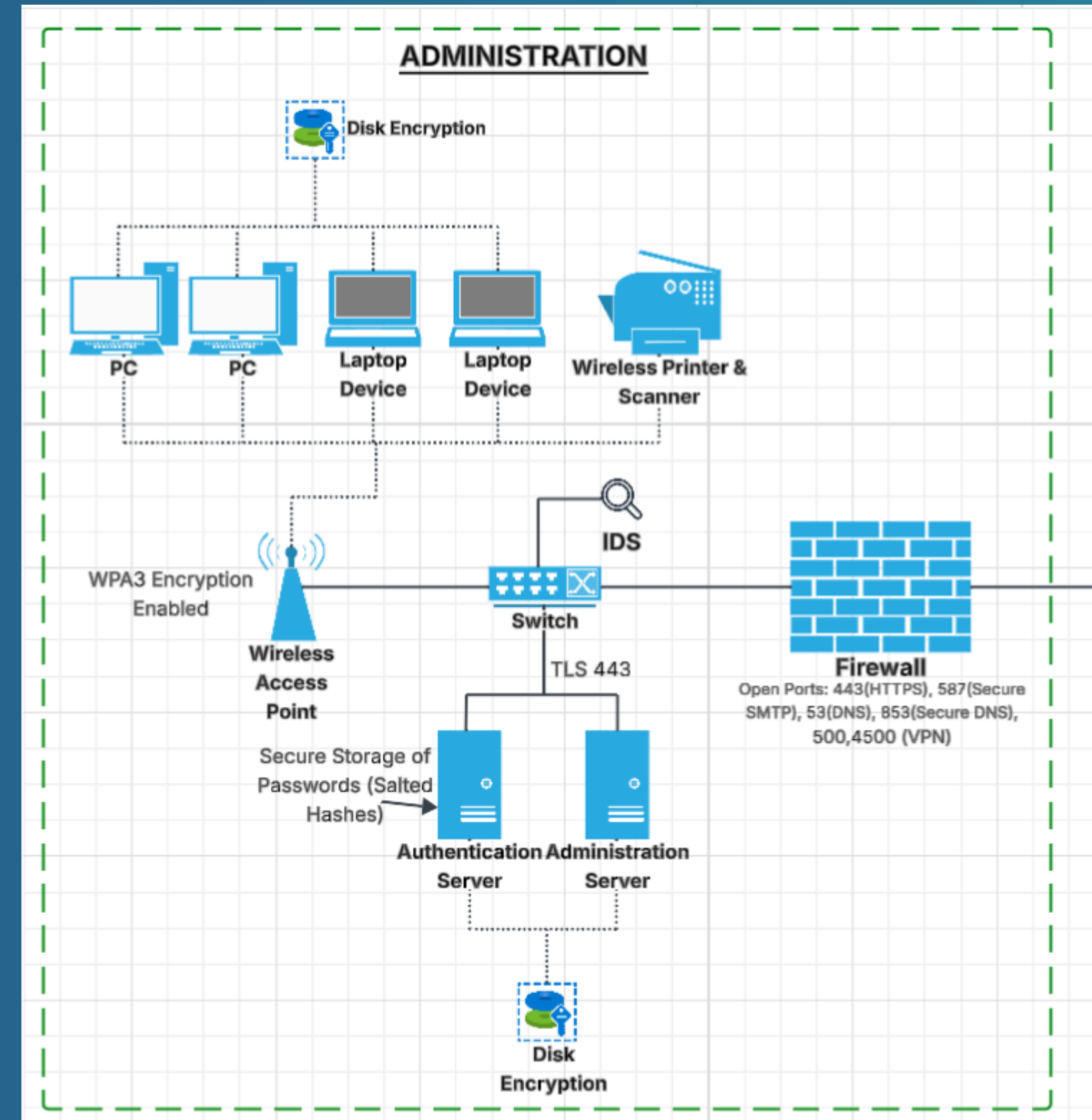
- This zone handles administrative tasks, employee services, authentication and Wi-Fi.
- Entities in this zone:
 - PCs & Laptops – Used by staff for administrative tasks such as HR, finance and reporting.
 - Wireless Printer & Scanner – Enables wireless printing and scanning for the employees in the office environment.
 - Authentication Server – Handles login credentials of users across the network and helps make sure that only authorized staff can access specific resources.
 - Administration Server – Hosts admin-related software and internal databases.
 - Wireless Access Point – Provides Wi-Fi connectivity for employee laptops and mobile devices.
 - Switch – Connects all internal devices and allows efficient communication between them.



Administration Zone



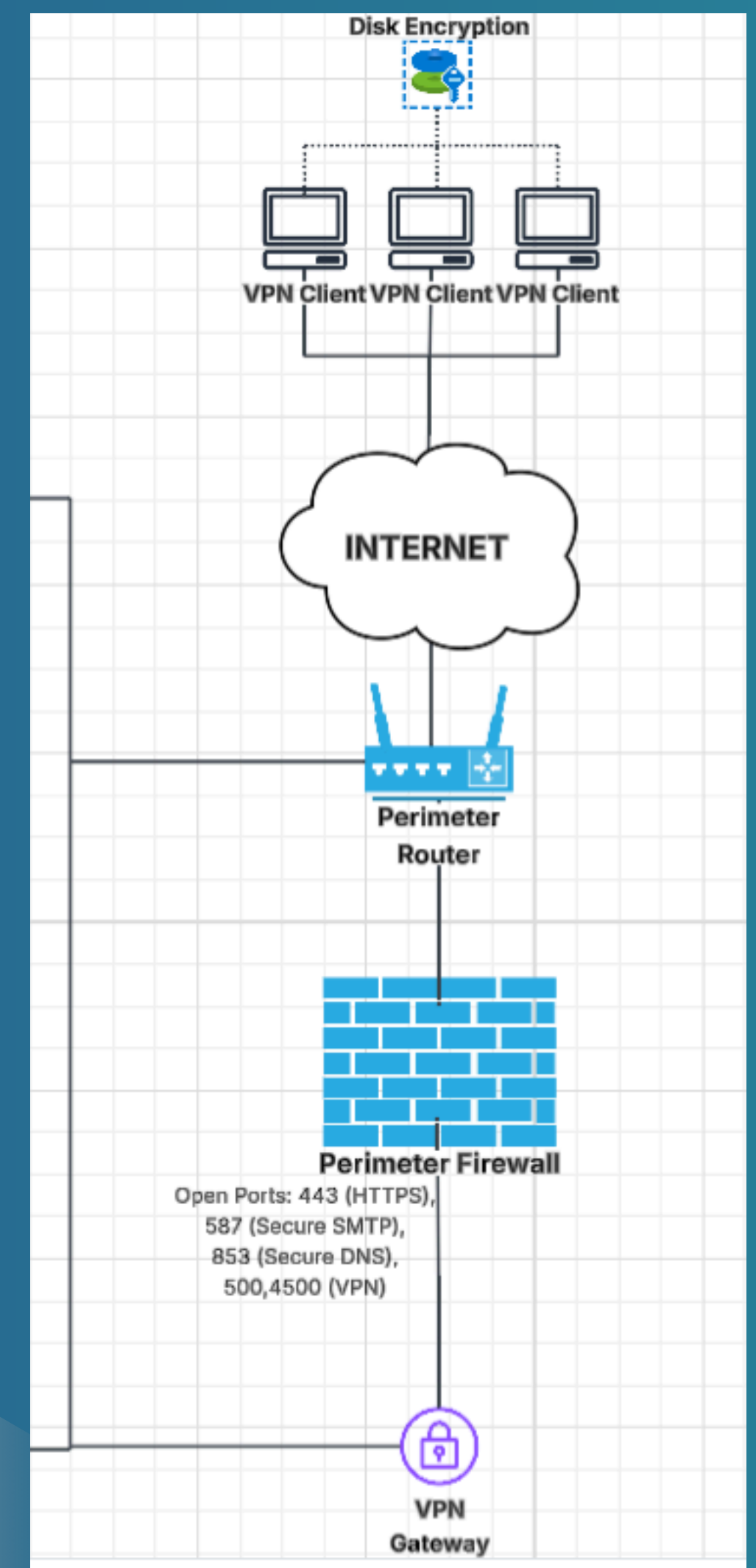
- Security controls in this zone:
 - Disk Encryption – Protects sensitive administrative data stored on PCs, laptops, and servers from getting stolen or leaked.
 - WPA3 Encryption – Used to protect Wi-Fi communication and keep sensitive information safe from unauthorized access.
 - IDS – Monitors internal traffic for suspicious behavior or threats and alerts to the network admins.
 - Firewall – Filters incoming/outgoing traffic to protect against unauthorized access and malware
 - Secure Storage Of Passwords (Salted Hashes) – Storing passwords using salted hashes significantly enhances security by protecting user credentials against brute-force and rainbow table attacks.



Perimeter and VPN Zone



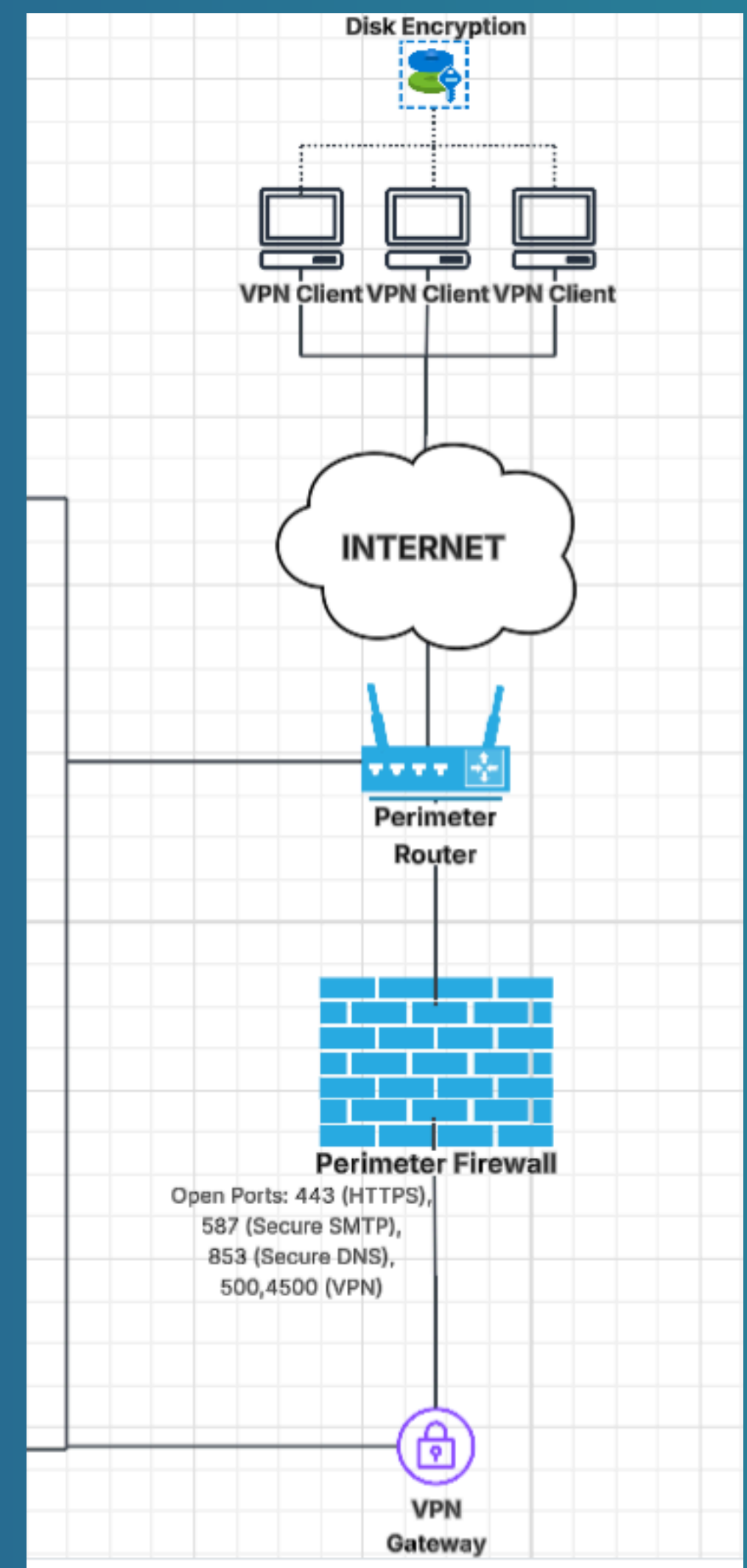
- This zone connects the internal network to the internet and provides secure remote access through VPN.
- Entities in this zone:
 - Internet – Provides external connectivity, allowing users to access online services and enabling communication between internal servers (like mail or web servers) and users outside the network.
 - Perimeter Router – Directs network traffic between the internal network and the internet, acting as the first line of control before traffic reaches the firewall.



Perimeter and VPN Zone



- Security controls in this zone:
 - Disk Encryption – Protects sensitive data stored on devices ensuring confidentiality even outside the internal network.
 - VPN Gateway – Creates secure, encrypted tunnels for remote users or sites to connect to the internal network.
 - VPN Clients – Remote devices that connect securely to the network using the VPN.
 - Perimeter Firewall – The first real line of defense that checks incoming traffic and blocks any unwanted or suspicious connections from getting into the network



Open Firewall Ports Per Zone



- **Production Zone firewall:**

- 443 (HTTPS) – Since data needs to be transferred frequently to the production PCs, production connects to the administration. Port 443 is open as it only allows secure web traffic into and out of the production zone.

- **Administration Zone firewall:**

- 443 (HTTPS) – Secure web access to admin tools, internal dashboards, or external services.
- 587 (Secure SMTP) – Secure sending of emails from internal admin clients to the mail server in the DMZ.
- 53 (DNS) – Translates domain names to IP addresses, allowing administrative devices to locate internal servers or permitted external services efficiently.
- 853 (Secure DNS) – Provides secure, encrypted DNS queries enhancing confidentiality and integrity of DNS traffic in the network.
- 4500, 500 (IPsec VPN) – Supports secure VPN access for remote employees. Ports 500 and 4500 are required for IPsec VPN.

Open Firewall Ports Per Zone



- **DMZ Zone Firewall:**

- 443 (HTTPS) – Secures web traffic to and from the public-facing web server, ensuring encrypted communication for websites and services accessible from outside
- 587 (SMTP with TLS) – Used for sending outgoing emails securely from mail clients to the mail server.
- 853 (DNS over TLS): Provides secure, encrypted DNS queries to protect domain name lookups from tampering.
- 25 (SMTP): Handles email delivery between mail servers, enabling the sending and receiving of emails across the internet.
- 53 (DNS): Used to translate website names into IP addresses so devices can find and connect to each other on the network.

Open Firewall Ports Per Zone



- **Perimeter Firewall:**

- 443 (HTTPS) – Enables secure encrypted communication for remote users accessing the VPN gateway.
- 587 (Secure SMTP) – Ensures secure email transmission when sending emails via the company's mail server.
- 853 (Secure DNS) – Provides secure and encrypted DNS lookup for VPN users, protecting their web browsing and preventing tampering of domain name requests.
- 4500, 500 (IPsec VPN) – Ensures secure remote access by establishing encrypted tunnels between external users and the company's network. Ports 500 and 4500 are required for IPsec VPN.

References



National Institute of Standards and Technology. (n.d.). NICE Framework Resource Center. U.S. Department of Commerce. Retrieved June 9, 2025, from <https://www.nist.gov/itl/applied-cybersecurity/nice/nice-framework-resource-center>

NSA Network Infrastructure Security Guide:

https://media.defense.gov/2022/Jun/15/2003018261/-1/-1/0/CTR_NSA_NETWORK_INFRASTRUCTURE_SECURITY_GUIDE_20220615.PDF

CWNP. (n.d.). What is a switch, router, gateway, subnet, firewall & DMZ? CWNP <https://www.cwnp.com/what-is-a-switch-router-gateway-subnet-firewall-dmz/>

[Tool Used: Lucid Chart](#)



THANK YOU!